



# Michael Boiman

Freelance Quality Engineer & AI Architect • 20+ Years QA • 4+ Years AI Development  
Frankfurt am Main, Germany · Freelance · available for new projects

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-80%  
**Inquiries**  
through quality monitoring

96%  
**Quality**  
after migration (was 30%)

80%  
**Auto-generated**  
AI-driven test cases

Quality Engineering | LLM/AI Architecture | CI/CD Automation | Multi-Cloud | Peppol/E-Invoice

## Career Profile

**Quality Engineer and AI Architect** with 20+ years in software quality assurance and 4 years of specialized AI development. What AI needs today — reliability, measurability, self-correction — I spent two decades building into systems; my focus: transforming business processes through measurable automation, from test generation to invoice processing.

In enterprise projects (DB Vertrieb, DVAG, TÜV Süd), I built quality monitoring systems, introduced AI-driven test automation, and secured legacy migrations with 24/7 validation.

Today I combine quality engineering with AI architecture: MCP and A2A protocols, multi-agent systems, e-invoicing platforms to EU standards, and a 40+ skill automation ecosystem for enterprise workflows.

I also pass this knowledge on — in developer and business workshops and as an invited guest speaker (TU Darmstadt) — translating hands-on AI for technical and non-technical audiences alike.

## Technical Skills

Quality Engineering | Playwright  
Cucumber/Gauge | Python  
LLMs (Claude/Gemini/GPT/Copilot)  
Agent Orchestration (MCP & A2A)  
Google ADK | LangChain  
Peppol/E-Invoicing (EN 16931)  
CI/CD | Kubernetes  
Azure Functions | Grafana  
Elasticsearch | REST APIs  
Docker | JMeter | Gatling

## Education

### Diploma in Computer Science (UAS)

Cologne University of Applied Sciences, Gummersbach  
2000 – 2005

### ISTQB Certified Tester 2005

### Certified Dynatrace Diagnostics Basic Training 2007

## Languages

German Native  
English Professional

## Additional Qualifications

ISTQB Certified  
Agile (Scrum/Kanban)  
4+ Years AI Development, 20+ Years Quality Engineering

## PROJECTS & SOLUTIONS

### Quality Dashboard - Real-time Overview

Grafana | Elasticsearch | Python | CI/CD | Playwright

Fully automated end-to-end solution for real-time quality monitoring across 10 test tools, 5 environments plus multi-source integration (Pipeline, LeanIX, Jira, Confluence) for automated PDF reports with open issues, bugs, and technical debt.

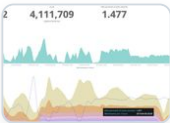


### 24/7 Automated Legacy Migration Validator: Dual-Run Quality Engineering

Java | Cucumber | Kibana | Graylog | Elasticsearch

#### Enterprise Legacy Transformation with Fully Automated Quality Assurance

**The Challenge:** Complex mainframe price calculation with millions of combination possibilities needed secure migration - without quality loss or data errors. **The Solution:** 24/7 Dual-Run Validation System with intelligent deviation analysis • **Continuous Testing:** Randomly generated tests across all variations around the clock • **Real-time Monitoring:** Live dashboard with immediate deviation detection...



### E-Invoicing Automation Platform

Python | Azure Functions | Peppol | Storecove | SAP ByDesign

**Fully Automated E-Invoice Processing to EU Standards Core Features:** • **Peppol Network Integration:** Connection to European e-invoicing network via Storecove Access Point • **Multi-Format Support:** ZUGFeRD, Factur-X, XRechnung, UBL 2.1, CII (EN 16931 compliant) • **ERP Integration:** Automatic transfer to SAP ByDesign with validation and error handling • **Hexagonal Architecture:** Domain-driven design with Ports & Adapters for maximum testability &...

## AI Context Orchestrator — Multi-Repo Development Platform

Python YAML Claude Code Plugins Hooks API GitHub Actions

**Central Control System for Enterprise AI Development Core Features:**

- **Context Orchestrator:** ecosystem.yaml as single source of truth for 20+ repos, customers, infrastructure and skills
- **40+ Production Skills:** Log analysis, deployment verification, incident handling, email management, meeting transcription, PDF/certificate generation, SharePoint integration
- **Hook-based Permission System:** Smart Guard with project-specific security rules (branch protection, secret detection, deployment gates)
- ...

## Email Classification and Processing Workflow

Python Azure Functions OpenAI GPT-3.5 Microsoft Graph API Elasticsearch

Complete automation solution with Microsoft Graph API, SAP RFC, Azure Functions (Python), OpenAI GPT-3.5-Turbo, Elasticsearch 8.x, Kibana Dashboards, Docker containers, and GitHub Actions CI/CD.



## Developer Workshop: AI-Agent Development & Integration

Anthropic Claude Google Gemini OpenAI GPT-4o MCP A2A

**Target Audience:** Developers, Technical Leads, DevOps Engineers

**Workshop Content:**

- **Hands-on AI Agent Development:** Live coding with Claude, Gemini, GPT-4o
- **Next-Gen Protocols:** MCP & A2A implementation for enterprise integration
- **Browser Automation:** Playwright-MCP for automated test generation
- **IDE Integration:** AI assistance in VS Code, Cursor, GitHub Copilot workflows
- **Multi-Agent Orchestration:** JSON-RPC, secure agent communication
- **Practical Demos:**...



## Professional Experience

### LLM Infrastructure Architect & Automation Engineer

06/2025 - present

#### BKS - AI Research & Development

**Development of LLM-orchestrated Enterprise Development Ecosystem:** Fully integrated system for knowledge management, project automation, and development workflows with Claude, Gemini, and OpenAI GPT.

#### Key Responsibilities

- **Open Source MCP Server Portfolio:** Development and publication of 3+ MCP servers on GitHub (bks-wiki-mcp, hubspot-mcp-bks, bks-codex) with uvx distribution, Claude Desktop integration, and multi-format support (Markdown, JSON, YAML)
- **Multi-Agent Orchestration System:** Implementation of production-ready A2A system with Orchestrator Agent (routing hub) and Knowledge Agent for inter-agent communication, capability-based Agent Cards, and JSON-RPC 2.0
- **Git Wiki Transformation:** Migration of Confluence knowledge base to structured Git-based wiki with hierarchical organization and multi-repository architecture (submodules for customer projects)
- **MCP Navigation Server:** Development of Model Context Protocol server (Python) with intelligent hierarchy navigation, automatic content discovery, and SharePoint integration
- **Google ADK Integration:** Implementation of Google Agent Development Kit for multi-agent systems and advanced tool integration

#### Tools & Technologies

- LLM Orchestration: Claude, Google Gemini, OpenAI GPT, GitHub Copilot, Model Context Protocol (MCP), Google Agent Development Kit (ADK)
- AI Development Platform: Claude Code, Plugin Architecture (Skills, Hooks, Commands, Agents), YAML/Markdown Configuration
- Multi-Agent: A2A Protocol, Orchestrator/Knowledge Agents, Agent Cards, JSON-RPC 2.0, SSE Streaming
- Backend: Python 3.13, FastAPI, uvx Distribution, Git Submodules
- Automation: GitHub Actions, GitHub GraphQL API, gh CLI
- Integration: SharePoint API, Clockify API, Elasticsearch, Natural Language Processing
- Development: Bash Scripting, jq, Docker, Multi-Agent Systems

## Quality Engineering

09/2025 - present

### TÜV Süd

**Quality Engineering** for an enterprise platform for AI-powered document processing in medical device certification (MDR/IVDR) with AI chat integration and data pipeline automation.

#### Key Responsibilities

- **BDD API Test Framework:** Development of comprehensive test framework with Python/Behave, self-healing authentication, HTML report generator with embedded API responses
- **E2E Test Automation:** Playwright E2E test suite for complete user journey (Setup -> Upload -> Transformation -> Download) with 500 error handling and route mocking
- **Performance & Monitoring:** Development of Azure Monitor Workbooks (QA Live Testing Companion, Error Investigation Assistant) with KQL for end-to-end monitoring
- **Bug Analysis & Quality Intelligence:** Implementation of comprehensive bug tracking system with WIQL queries, automated dashboards, and executive PDF reporting
- **Pipeline Integration:** Test automation in Azure DevOps CI/CD pipelines with HTML report upload to Azure Test Results

#### Tools & Technologies

- Testing: Behave, Playwright, TypeScript, Python, pytest, Page Object Model, Fixtures
- Monitoring: Azure Monitor Workbooks, KQL, Application Insights, Log Analytics
- Backend: FastAPI, Python, Pydantic, SQLAlchemy, Azure Functions, Azure Service Bus
- Frontend: React, TypeScript, Vite, TanStack Router
- AI Integration: Azure OpenAI (GPT-4o), LangChain, Azure AI Search, Embeddings
- DevOps: Azure DevOps Pipelines, Git, Docker, Azure Blob Storage, Poetry

## AI-driven Automated QA Environment for Energy Infrastructure

07/2024 - 01/2025

### AkkuSwap Startup

**QA Leadership for EU-wide Battery Swap Infrastructure:** AI-driven simulation tool for battery swap station network with energy infrastructure integration

#### Key Responsibilities

- QA concept, design and implementation for AI-driven simulation of EU-wide infrastructure
- QA framework development for inhouse AI server infrastructure
- Proof-of-concept for automated testing of AI-driven simulations

#### Energy Sector Connection

- **50 Hertz Participation:** 50 Hertz (Elia Group subsidiary) participated in research project eHaul, the predecessor of AkkuSwap
- Direct energy sector experience with critical infrastructure quality requirements

#### Technical Stack

- Infrastructure: Linux, Docker, Azure OpenAI
- Automation: pytest, AI code generation integration patterns
- Monitoring: Grafana, Azure Monitoring
- Energy Systems: Battery swap infrastructure, grid integration simulation

#### Key Achievements

- Established QA methodology for AI-driven infrastructure simulation
- Created testing framework for energy sector critical systems
- Validated proof-of-concept with measurable reliability improvements

## Technical Lead & AI Automation Architect

01/2024 - 04/2025

### BKS

**Technical lead & development:** Design and implementation of a scalable automation solution for processing e-invoices and PDF invoices fetched from a mailbox of a leading online job platform (under NDA). Full responsibility from requirements analysis through implementation to production go-live.

#### Focus areas:

- **Workflow automation:** Extraction, analysis and classification of incoming invoices using Python and Azure Functions.
- **E-invoice classification & data extraction:** Full processing in accordance with EU standard EN 16931 (ZUGFeRD, XRechnung, Factur-X), including structural and content parsing.
- **Peppol Network Integration:** Connection to European e-invoicing network via Storecove Access Point for automated invoice receipt and dispatch with UBL/CII formats.
- **SAP ByDesign Integration:** Automatic transfer of validated invoice data to SAP ByDesign for seamless ERP integration.
- **Hexagonal Architecture:** Domain-driven design with Ports & Adapters for maximum testability and interchangeability of infrastructure components.

#### Tools & technologies

- Azure Functions: Serverless orchestration of automation workflows.
- Python: Core logic for data extraction, classification and integration with automated CI/CD processes, packaging, and comprehensive unit and integration tests for quality assurance.
- E-Invoicing: Peppol Network, Storecove API, ZUGFeRD/Factur-X, XRechnung, UBL 2.1, CII
- ERP Integration: SAP ByDesign, Salesforce
- Microsoft Graph API: Access to Outlook mailboxes and SharePoint libraries.
- GitHub: Version control, CI/CD pipelines and collaborative development.
- Kibana: Monitoring, log analysis and performance dashboards.
- Streamlit: For developing an interactive web application for visualization.
- AI Automation: Claude AI, Custom Skills Development, Git Analysis, Natural Language Processing
- Protocols: Model Context Protocol (MCP), Agent-to-Agent (A2A), JSON-RPC 2.0

## AI Engineering Lead & ML Solutions Architect

04/2023 - 04/2024

### BKS on behalf of Ryze

**Project management & technical lead:** Development and deployment of an AI-driven email bot to automate customer communication, AI-assisted processing of incoming emails and triggering of follow-up activities in existing SAP systems. Responsible for end-to-end project coordination and team management from idea to implementation."

#### Focus areas:

- **Development of ML models:** Design and training of specialized models for effective email classification, implemented via Azure Functions.
- **API integrations:** Seamless connection to Microsoft Outlook 365 via Microsoft Graph API and to SAP systems for customer data checks.
- **Process automation:** Automatic moving of emails based on classification and specific business rules, including complaint handling.
- **DevOps and monitoring:** Implementation of CI/CD pipelines and use of Kibana for monitoring.
- **Development of a Streamlit-based user interface:** Improved testability and visualization of process results.

#### Tools & technologies

- Azure Functions: For serverless application architectures and ML model execution.
- Python: Main programming language for the email bot and ML models.
- Microsoft Graph API: For accessing and processing emails in Microsoft Outlook 365.
- SAP API: For querying customer data and communicating with SAP systems.
- GitHub: For version control, CI/CD pipelines and collaborative development.
- Kibana: For real-time monitoring and system performance analysis.
- Streamlit: For developing an interactive web application for system monitoring.

## Platform Quality Architect (CI/CD, Monitoring, Automation)

08/2021 - 05/2025

### DVAG

**Test management & coordination,** as well as introduction of CI/CD with quality-gate setup. Acting as cross-team Quality Lead: onboarding, guidance, consulting and training for the introduction of quality metrics and test automation within the pipeline across teams.

#### Tools

- IntelliJ / Visual Studio Code
- Java / JavaScript / TypeScript
- Playwright / Gauge / Karate
- Python for processing and analysis of test and reporting data
- CI/CD / GitHub - reusable workflows
- Helm / Containers
- JIRA / Confluence
- Grafana
- Behaviour-Driven Development
- Consumer-Driven Contracts (Pact.io)

## DevOps Quality Lead & Dashboard Architect

01/2017 - 05/2021

### DB Vertrieb

**Test management, test coordination, test automation - agile (Kanban, Scrum, SAFe):** Coordination as PO of a cross-team QA team including cross-team onboarding, support, consulting, training

#### Tools

- IntelliJ
- Java
- Cucumber / Cypress
- JMeter / Gatling
- CI/CD / GitLab CI / Jenkins
- Helm / Docker
- JIRA / Confluence
- Kibana / Grafana / Instana / Graylog
- Behaviour Driven Development
- Consumer Driven Contracts (Spring Cloud Contract)
- Ionic (hybrid app)
- Selenium

## Earlier Positions

|                   |                                                                                                                                                                                     |                                                                                |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 09/2015 - 01/2017 | <b>Performance Engineering Lead &amp; Test Architect</b> — DB Systel                                                                                                                | Load and performance tests and analysis - test manager, test designer, analyst |
| 04/2015 - 09/2015 | <b>Senior iOS Developer &amp; Mobile Architect</b> — Telekom                                                                                                                        | Senior iOS Developer                                                           |
| 05/2009 - 03/2015 | <b>Quality Assurance Lead &amp; Test Automation Architect</b> — Siemens                                                                                                             | QA, test management, test automation, requirements engineering                 |
| 03/2009 - 05/2009 | <b>Performance Test Engineer &amp; Load Testing Specialist</b> — ING-DIBA                                                                                                           | Load and performance tester                                                    |
| 01/2006 - 12/2008 | <b>Performance Test Engineer &amp; Quality Consultant</b> — British Telecom, Mobiliar, DB-Systel, Sparkassen Informatik, Loyalty Partner, Telekom, Itelium, Deutsche Post, Postbank | Load and performance tester / tester                                           |

# Workshops & Presentations

**Guest Lecture: AI-Assisted Software Development in Practice**  
TU Darmstadt — Information Systems Group · Lab course “AI Startup: From Idea to Execution” — 04/2026

**Workshop: Development with Generative Language Models**  
Developer Workshop - Agent-based Software Development — 06/2025

**Presentation: Efficient Documentation through Automation**  
Enterprise Presentation - AI-powered Documentation Workflows — 04/2025

**AI Workshop for Business Integration**  
Applications of AI in Business Context — 2023